Appendix for CW2 Report

Spatial Variation in Airbnb Listing Counts across London

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Appendix A. Code

Code and outputs are available in the public GitHub repository Shui-Zhou/spatial-data-analysis-final (https://github.com/Shui-Zhou/spatial-data-analysis-final).

Main analysis scripts:

- Analysis/step1_eda_official_tfl.py
- Analysis/step2_ols_baseline.py
- Analysis/step3_spatial_autocorrelation.py
- Analysis/step4_gwr_models.py
- Analysis/step5_spatial_regression_robustness.py

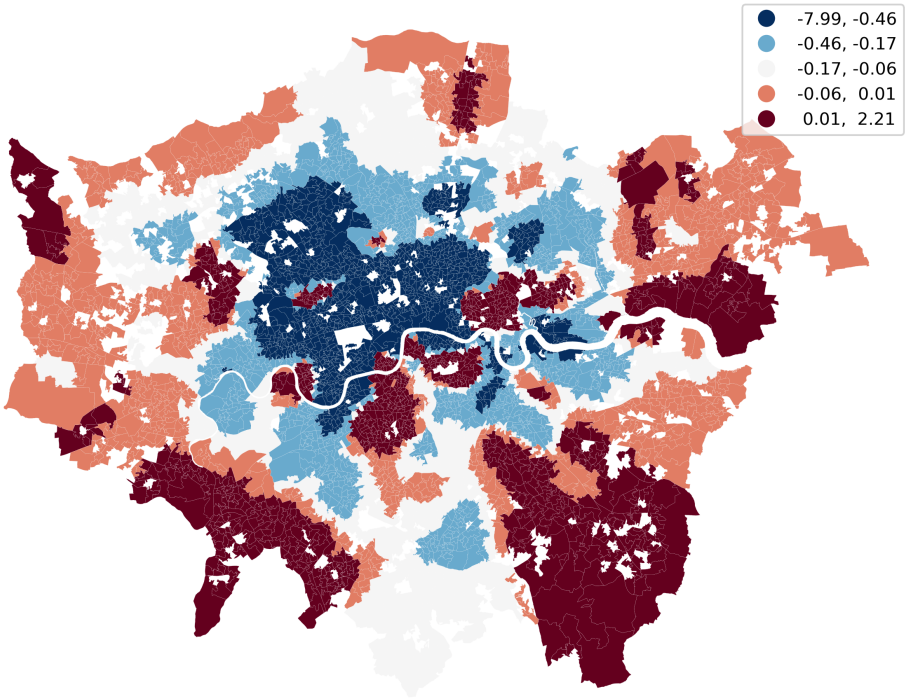
Environment:

- mscui2026 conda environment
- Python 3.12 with geopandas, mgwr, libpysal, and spreg

Appendix B. Robustness Checks

Spatial Lag and Spatial Error models (Step 5) confirm strong global spatial dependence (rho = 0.715, lambda = 0.733) but do not capture local variation. Post-GWR residual Moran's I falls sharply relative to OLS, dropping from 0.556 to 0.0668 in the raw-count GWR and 0.0777 in the log model, although some residual dependence remains. The log-transformed sensitivity model shows higher adjusted R^2 and fewer effective parameters, while the raw-count GWR leaves slightly less residual spatial autocorrelation. OSM-derived station features (n = 307) were also tested against the official TfL data (n = 385), with directionally consistent results across model specifications.

Appendix Figure 1 supplements the discussion in Section 3.5 by visualising the local coefficient surface for rapid-transit distance. More negative values indicate places where proximity to a station is associated with higher Airbnb counts, which is most evident in parts of north and west London.

Appendix Figure 1. Local GWR coefficient for rapid-transit distance in the raw-count model.

Local GWR coefficient for rapid-transit distance